

Open Resources for Corpus-Based Learning of Ancient Greek in Persian

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Abstract

This paper illustrates a new approach to learning historical languages, focusing, in this case, on learning ancient Greek. Instead of manual translation, we used treebanks data and localized grammatical explanations, and utilized machine translation to make commentaries and focus on the precise vocabulary, morphology, and syntax to understand. We report on how, after thirty hours of instruction, Persian speakers were able to critique existing Persian translations of the *Iliad* and create more accurate representations of the Greek original. The same learners were, at this point, able to begin producing the first direct translations of Plato's *Crito* into Persian. As a result, we have three finalized translations of *Crito* under development that each represent a different interpretation or approach to the text. All of the resources used in the courses, including treebanks, translations, textbooks, and grammars, are openly licensed or in the public domain. Our *Crito* translation will also be released under a Creative Commons License.

Keywords: Ancient Greek; parallel corpora; translation alignment; dependency trees; historical languages.

Introduction

In recent years, there has been an increasing awareness of the use of technology as a pedagogical tool that can be used to accelerate the development of educational resources in different languages, particularly low-resource ones. In this paper, we have focused on using cross-lingual digital annotation to develop pedagogical resources in Persian for learning ancient Greek.

Although Greek language and culture have a well-established influence in the Persian-speaking world through centuries of cultural exchange, the ancient Greek language has not been studied extensively, if at all, in Persian-speaking academic institutions—there are, for example, no regular courses on ancient Greek in any Iranian universities. Classes occasionally take place, but on an ad hoc and normally informal basis. Insofar as translations of Ancient Greek sources exist in Persian, the vast majority have not been translated from Greek to Persian directly, but indirectly through English, French, or German. Many texts have not only one, but multiple indirect translations. Such indirect translations lead to inaccuracies and misinterpretations that have, in turn, had a fundamental effect on how ancient Greek texts have been received by Persian speakers (Palladino, Shamsian, and Yousef 2022).

Considering the historical and intellectual significance of ancient Greek sources in Persian-speaking countries, having access to the original Greek texts instead of depending on indirect translation is crucial. In the case of this paper, we have focused on resources for the Homeric epics. However, the methods could be expanded to other Greek texts or even other languages.

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Digital tools and annotations have the potential to facilitate the development of resources for low-resource languages to a great extent (Crane 2019). Hundreds of books and papers have been written about the Homeric epics in English; however, it would be extremely challenging to access most of them in another language. Even if they were in the public domain or openly licensed, manual translation of this much text into another language is time consuming and costly. However, some of these sources could become much more accessible in other languages through machine translation, given that machine translation was available for the target language. We have used machine translation to make lexicons, commentaries, and grammars more accessible. Nonetheless, this is only applicable to texts that are already machine readable and available under an open license.

Given the challenges of translating traditional sources about Greek into a language such as Persian, we decided to concentrate on materials that may be localized rather easily and are largely independent of learners' first language (L1).

More than one million words of ancient Greek are available in treebanks—textual databases in which annotations identify the part of speech, standardized dictionary form, and syntactic role of each word in a sentence (de Marneffe et al. 2021). A corpus of a million words is larger than the assigned reading list for any PhD program in Ancient Greek with which we are familiar—it provides far more reading materials than anyone but the most ambitious of learners could cover without years of work.

Treebanks have shown to be an effective pedagogical tool in the classroom for teaching ancient Greek in English, encouraging learners to engage more closely with the original text (Gorman 2020; Mambrini 2016). These treebanks use consistent tags for describing Ancient Greek and by learning how to understand and use dependency trees, learners, regardless of their first language(s), can interact with the text more actively. Most beginner textbooks of Ancient Greek use simplified, easier to explain versions of more complex authentic texts. However, using annotations empowers learners to read authentic Greek text from the very beginning.

Although having cross-lingual annotations gives us numerous opportunities, we still need grammar manuals and textbooks, and the problem of localizability applies to these as well. The examples, explanations, and even exercises in such books are often written for English speakers. For instance, authors of a textbook about ancient Greek in English may not consider it necessary to include proper explanations of definite articles because the concept is already familiar to an English speaker. Considering that Persian does not have definite articles, Persian speakers may require more explanation of what a definite article is and how it is used. To address this issue, we created an open-source, localizable reference for learning Greek based on Smyth's *Greek Grammar*, designed to include explanations for each concept, regardless of how familiar or unfamiliar the subject is for speakers of a different language. Explanations that could not be translated well into Persian were edited, and the text was simplified to get better results from machine translation. This localizable reference, called the [Didakta Grammar for Annotation](#), provides explanations of syntactic complexities that are not represented in treebanks. Treebanks can tell learners that a word is in the dative case and modifies a verb, but additional grammatical explanations can tell learners that the dative describes an instrument (e.g., “the hero was struck with a spear”) vs. a location (e.g., “the hero was struck on the shoulder”).

In addition to treebanks and the *Didakta Grammar for Annotation*, we adopted a technique of annotating and aligning translations with the original ancient Greek at word level as a pedagogical tool. We integrated [Ugarit](#), an open-access digital alignment tool, for learning and assessment. Ugarit is a text alignment tool that allows you to manually annotate parallel text and generate word-level translation alignment in a variety of languages (Yousef et al. 2022). As a vocabulary learning tool, translation alignment provides context for each word and increases the chance of incidental learning (Foradi 2020). Conventionally, students of ancient Greek are asked to translate a Greek text into their first language or compose a text in Greek to assess their knowledge of the language. The assessment of their translations is, among other issues, a time-consuming task for teachers and cannot easily be automated. Meanwhile, translation alignment along with multiple-choice questions can be used to assess a learner's understanding of an ancient Greek text. The knowledge of more advanced learners could also be assessed by their ability to create treebanks for complex sentences; however, we found this to be too complex for beginners.

An Introductory Course in Homeric Greek

Here, we report on the content and results of an introductory online course on Homeric Greek for Persian speakers taught by Farnoosh Shamsian, running from July to December 2021. The course included thirty sessions, each ninety minutes, and was taught in Persian to twenty-four initial participants with no prior knowledge of ancient Greek. The goal of this course

was to prepare learners for reading Homeric poems in ancient Greek with the help of lexicons and grammar books. Students in this course, unlike most Greek courses, were never assigned to compose in Greek and instead, we focused solely on reading authentic texts. There is no textbook for Homeric Greek available in Persian and this course was based completely on open digital resources, including the [Perseus Treebank](#) (Crane et al. 2018), public domain sources such as [Pharr's Homeric Greek](#) (1920), and a born-digital Persian translation of book one of the *Iliad* aligned with the original Greek.

The advantages of using corpora in language learning are well-established (Vyatkina and Boulton 2017); however, it is still not widely applied to learning historical languages. The syllabus for this course was based on frequency data derived from the annotated corpus of the *Iliad*, giving priority to the most frequent inflections and syntactic structures. Considering that many textbooks of ancient Greek are not arranged based on the frequency of morphological and syntactic structures, the sequence of subjects in the Homeric Greek course was not very conventional. For instance, textbooks often introduce the present tense prior to the imperfect or the aorist tense; however, frequency data clearly shows that the present tense is not the most frequent tense in the *Iliad*. According to treebanks data, aorist indicative active tense was used 3948 times in the whole *Iliad*, while there were only 1469 instances of present indicative active tense. Therefore, aorist indicative active verbs were discussed in the seventh session of the Homeric Greek course and present indicative active verbs in the fifteenth session. The imperfect indicative active tense, with 2669 instances in the whole *Iliad*, was discussed in the ninth session.

Given that both Persian translations of the *Iliad* are derived from mediating French translations, it is not surprising that neither of them aligns well with the original Greek text. Therefore, book one of the *Iliad* was translated to Persian and aligned at word level. This aligned direct translation was used to help learners gain a better understanding of its syntax and navigate through the corpus more efficiently. The goal was to use parallel corpora instead of or in addition to traditional reference tools such as dictionaries, an approach that has already been shown to improve the translation quality for modern languages (Liu 2020).

To complement treebanks data, a grammar textbook is also crucial for providing adequate information on morphology and syntax. Recent approaches to a digital Greek grammar references give more freedom to the reader than conventional print books do. For instance, *Overview of Greek Syntax* by Rydberg-Cox, openly accessible in Perseus Digital Library, arranges the content of Smyth's (1916) reference grammar in an innovative way (Rydberg-Cox 2000). More recently, the Pedalion modular syntax project developed at KU Leuven enables the reader to access information flexibly compared to more traditional printed grammar books that often have a fixed structure (Van Hal and Keersmakers 2021). With such a flexible approach to grammar referencing, we can take advantage of the benefits of digital media more effectively.

The main difficulty with using English-language Greek grammar books and textbooks with Persian speakers is that some of the explanations and examples are written exclusively for English speakers and thus cannot be translated; some textbooks even assume prior knowledge of Latin. To provide a localizable reference book, a modular grammar based on Smyth's *A Greek Grammar for Colleges* and inspired by the work of Rydberg-Cox and Pedalion was developed by the instructor. In this modular grammar, Didakta Grammar for Annotation, the goal was to keep the explanation as localizable as possible while avoiding complex language that would cause inaccuracies in machine translation. The same approach was taken for developing a textbook based on Pharr's *Homeric Greek* (1920). Both the grammar reference and the textbook have already been translated into Persian and are openly accessible.

Additionally, Ugarit was used in this course, both as an assessment tool by the instructor and as a vocabulary learning tool by learners, with which they could review each word in its context. Moreover, the process of aligning a translation in itself helps learners to develop a better understanding of the text semantically through systematic comparison (Palladino 2020).

Hom. I. 1. 43-54

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Ελληνική Transliterate

Transliterate فارسی

ὥς ἔφατ' εὐχόμενος , τοῦ δ' ἔκλυε Φοῖβος Ἀπόλλων ,
 βῆ δὲ κατ' Οὐλύμποιο καρήνων χωόμενος κῆρ ,
 τόξ' ὤμοισιν ἔχων ἀμφηρεφέα τε φαρέτρην :
 ἔκλαξαν δ' ἄρ' οἴστοι ἐπ' ὤμων χωόμενιο ,
 αὐτοῦ κινήεντος : ὃ δ' ἦε νυκτὶ ἑοικώς .
 ἔζेत' ἔπειτ' ἀπάνευθε νεῶν , μετὰ δ' ἰὸν ἔηκε :
 δεινὴ δὲ κλαγγὴ γένετ' ἀργυρέοιο βιοῖο :
 οὐρῆας μὲν πρῶτον ἐπώχετο καὶ κύνας ἀργούς ,
 αὐτὰρ ἔπειτ' αὐτοῖσι βέλος ἔχεπευκὲς ἐφίεις
 βάλλ' : αἰεὶ δὲ πυραὶ νεκύων καίοντο θαμειαί .
 ἐννήμαρ μὲν ἀνὰ στρατὸν ὥχετο κῆλα θεοῖο ,
 τῇ δεκάτῃ δ' ἀγορὴν δὲ καλέσασαο λαὸν Ἀχιλλεύς :

دعاکنان چنین می‌گفت و آپولون فویبوس او را می‌شنید . با دلی خشمگین , کمان و
 تیردان دوسرپوشیده بر دوش گرفته , از بلندای المپ پایین آمد ; هنگامی که خود
 حرکت کرد تیرها بر شانه‌های آن خشمگین [به هم خورده] به صدا درآمدند و او
 همچون شب می‌آمد . آنگاه دور از ناوها نشست , و تیری به میان انداخت ; بانگ
 کمان سیمین , سه‌گین بود . نخست پی قاطران و سگان چابک می‌تاخت اما سپس
 تیری بُرنده سوی خود آنان انداخت و به هدف می‌زد , و هیمة‌های آتش اجساد ,
 پرازدحام , بی‌وقفه می‌سوختند .
 خدنگ‌های خداوند نه روز بر فراز لشکر می‌رفت و در [روز] دهم آخیلئوس مردم
 را به مجمع فرا خواند .

(8) (91) 92% GRC - FAR

(116) 94% GRC - FAR

(7)

Figure 1. Word-level alignment of the *Iliad* 1.43-54 done by a learner in the Homeric Greek course.

Translating Homeric Greek to Persian

Since both available translations of the *Iliad* in Persian are indirect, they often do not match up well with the original Greek text in a word-level alignment, leaving multiple tokens unaligned either in the original text or in the translation itself. The learners in the Homeric Greek course, however, were able to provide more accurate translations by using digital annotation. For instance, after less than twenty hours of instruction, learners were asked to revise the indirect Persian translations of the following lines:

[4.3] τοὶ δὲ χρυσέοις δεπάεσσι [4.4] δειδέχατ' ἀλλήλους, Τρώων πόλιν εἰσορόωντες: [4.5] αὐτίκ' ἐπειράτο Κρονίδης ἐρεθιζέμεν Ἥρην [4.6] κερτομίοις ἐπέεσσι παραβλήδην ἀγορεύων: (Hom. Il. 4. 3-6)

A.T. Murray's translation (rev. Crane):

and they with golden goblets toasted one the other as they looked upon the city of the Trojans. And right away the son of Cronos began trying to provoke Hera with mocking words, saying with malice:

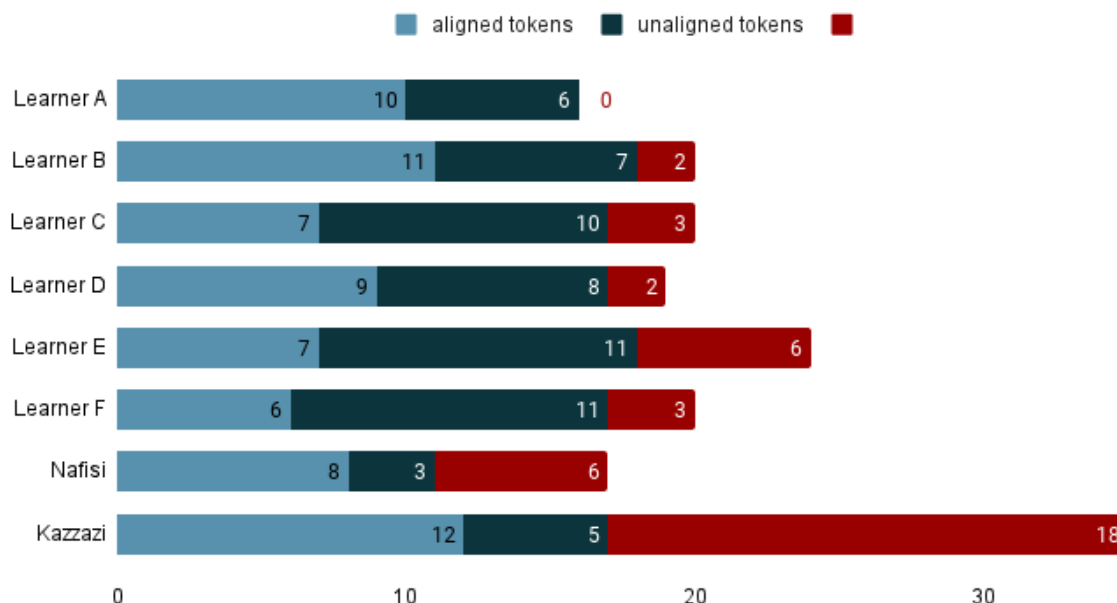


Figure 2. Translation pair ratios across different translations of Hom. Il. 4. 3–6.

Students' translations were compared to those of other translators using translation alignment. As we see in the graph, the indirect translations leave a higher number of unaligned words in comparison to the students' translations. Kazzazi's translation includes extra synonyms and explanations that do not exist in the original Greek and are left unaligned. However, this graph only shows the unaligned tokens in the Persian translation and does not illustrate the unaligned words in the Greek text, where we see that Nafisi's translation leaves six words unaligned, while some other translations have only one or two unaligned tokens in Greek, and two of them (learners B and E) leave no Greek word unaligned. All alignments were done by one annotator following the same guidelines. More details can be seen in the following word-by-word glosses. For styling reasons, Persian texts have been replaced by transliterations.

Nafisi's translation (1958, 143)

([View the alignment on Ugarit.](#))

Transliteration and word-by-word glosses for Nafisi's translation:

va [and] ānhā [they] sāqarhāye[goblets] zarrin [golden] be [to] dast [hand] čašmān [eyes] rā [o]¹ bar [on] divārkhāye [walls] tro(v)ā [troy] duxte budand [gazed (literally, sewed)]. xodāye [god of] xodāyān [gods] ke [that] mixāst [wanted] herā [Hera] rā [o]¹ be xašm āvarad [provoke] čonin [such] goft [said]

Revised machine translation results:

And they were staring at the walls of Troy with golden goblets in their hands. The god of the gods, who wanted to anger Hera, said:

Kazzazi’s translation (1998, 93)

([View the alignment on Ugarit.](#))

Transliteration and word-by-word glosses for Kazzazi’s translation:

ānān [they] jāmhāye [goblets] zarrin [golden] rā [o]¹ be [to] ham [each other] midādand [gave] va [and] dar [in] hamān [same] hengām [moment] bar [on] šahre [city of] tro(v)āieān [trojans] minegaristand [were looking]. pas [then] kronosi [of Kronos] xāst [wanted] ke [that] bekušad [try] tā [to] bā [with] soxanāni [words] sard [cold] va [and] delāzar [upsetting] herā [Hera] rā [o]¹ barāšubad [irritate] va [and] be xāšm avarad [provoke]; az in ruy [therefore], faribkārāne [deceitfully], vey [her] rā [o]¹ goft [said]:

Revised machine translation results:

They gave each other the golden goblets and at the same time were looking at the city of Trojans. So [son] of Kronos wanted to try to irritate and provoke Hera with cold and upsetting words; Hence, he deceitfully said to her:

Student A’s translation

([View the alignment on Ugarit.](#))

va [and] jāmhāye [goblets] zarrin [golden] bargerefte [taking up] bar [on] farāze [over] šahre [city of] tro(v)a [Troy] čāšm [eye] dāšand [have]. pure [son of] kronos [Kronos] ke [that] dar [in] talāš [effort] barāye [for] xāšmgin [angery] sāxtane [to make] herā [Hera] bud[was], biderang [immediately] bā [with] soxanāni [words] gazande [harsh] faryād barāvard [shouted]

Revised machine translation results:

And they sat their eyes on the city of Troy, taking up their golden goblets. Son of Kronos that was trying to anger Hera, immediately shouted with harsh words

Student B’s translation

([View the alignment on Ugarit.](#))

Transliteration and word-by-word glosses:

va [and] ānān [they] hengāmi ke [when] šahre [city of] tro(v)ā [Troy] ra [o]¹ minegaristand [were watching] moteqābelan [together] jāmhāye [goblets] zarrin [golden] boland kardand [lifted up]. pesare [son of] kronos [Kronos] biderang [immediately] qasde [attempt] āzordane [to annoy] herā [Hera] kard [did] va[and] be amd [deliberately] bā [with] kalamāti [words] gazande [stinging] soxan rānd [made a speech]

Revised machine translation results:

And they lifted up their golden goblets together as they were watching the city of Troy. Son of Kronos immediately attempted to annoy Hera and deliberately made a speech with harsh words

We have restricted our transliterations in this paper to the first two examples, however, the remaining alignment data is available on Ugarit:

- [Student C](#)
- [Student D](#)
- [Student E](#)
- [Student F](#)

The frequent use of participles and infinitives in ancient Greek is one of many ways in which it differs significantly from Persian. Furthermore, there are no definite articles, optative mood, middle voice, grammatical cases, or gender in Persian. Many learners often find such linguistic differences difficult to understand and render in a translation. For instance, participles and infinitives are commonly translated to finite verbs in Persian; however, we can see that many learners were successful in translating participles and infinitives. The majority of them (students A, B, C, and F) translated ἐρεθίζεμεν, which is an infinitive in Greek, to an infinitive in Persian. Another case in point is εἰσορόωντες, a Greek participle that students A and D were able to translate as a participle in Persian. Nonetheless, in many cases, we don't see such flexibility of direct options in the target language. In the same sentence, the Greek participle ἀγορεύων could not be translated into a participle in any of the translations due to the linguistic limitations of the target language.

We see innovative attempts by learners to translate certain complexities of the Greek language to Persian more accurately. For instance, παραβλήδην, an adverb coming from the verb παραβάλλω, is accordingly translated to an adverb from a verbal root by Student F (“ta’ne zanān,” from the verb “ta’ne zadan”).

Having access to the original text allowed the learners to read various lexicon entries for specific words that had complex meanings and even look through commentaries to deepen their understanding of those words. For instance, in session eighteen of the Homeric course, the learners were asked to translate the following sentence:

ὥς τὸν μὲν λίπε θυμός, ἐπ’ αὐτῷ δ’ ἔργον ἐτύχθη ἀργαλέον Τρώων καὶ Ἀχαιῶν: οἱ δὲ λύκοι ὥς ἀλλήλοισι ἐπόρουσαν, ἀνὴρ δ’ ἄνδρ’ ἐδνοπάλιζεν. (Hom. Il. 4.470–472)

A.T. Murray's translation (rev. Crane):

So his spirit left him, and over his body was wrought grievous toil of Trojans and Achaeans. Even as wolves they leapt upon each other, and man made man to reel.

In this sentence, ἐδνοπάλιζεν is an obscure word that only occurs once in the whole *Iliad*; therefore, understanding it without the help of commentaries would be difficult. Being able to openly access commentaries as machine-readable text in these situations was especially helpful because even Persian-speaking learners who did not know English well could use machine translation to better understand the Greek text and provide a more accurate translation. In the translation of ἀνὴρ δ' ἄνδρ' ἐδνοπάλιζεν, the learners had access to two commentaries on the Perseus Digital Library, one of which also points out that “ἀνὴρ ἄνδρα” [man(sbj) man(obj)] is a poetic rendering of “ἀλλήλους” [each other]. Some examples from the translation of the phrase are given below:

- Student A: “and each warrior pounded an opponent to the ground”
- Student B: “and (they) jostled each other”
- Student C: “and each agitated the other”
- Student D: “a man struck a man to the ground”

Nevertheless, depending on how complicated the text or annotations are, it might occasionally be difficult to find the correct information. Novice learners might need practice to learn to navigate through long and complicated lexicon entries. If machine translation is used, there is also a chance of getting inaccurate results that would mislead learners. One case in point is the translation of δειδέχαι ἀλλήλους in the aforementioned exercise from session twelve by Student C as “they made a pact with one another.” This is an illustration of a situation where a learner may not have fully understood the commentaries as a result of inaccurate machine translation, though these occurrences have been rare. Bearing the benefits of having access to such a vast range of information in mind, we consider such errors to be rather negligible.

The alignments for the translations of Hom. Il. 4.470–472 are available on Ugarit:

- [Student A](#)
- [Student B](#)
- [Student C](#)
- [Student D](#)

Translation of *Crito*

In the spring of 2022, seven participants of the Homeric Greek course formed a translation group to produce the first direct translation of Plato's *Crito* under the supervision of Farnoosh Shamsian. While *Crito* has four Persian translations, none of them are derived directly from the original Greek. Working with treebanks, commentaries, lexicon entries, and indirect Persian translations, each participant either refined an indirect Persian translation of *Crito* or translated it once again into Persian. All Persian translations were aligned to the Greek text by the translators themselves. The group met on a weekly basis to read and compare the translations. The initial goal of the group was to provide one finalized translation of *Crito* to Persian, but considering the high quality of the translations and the complexity of the text, it was decided to provide three finalized translations instead of one.

During the weekly meetings, the participants were able to not only critique existing Persian translations, but to actively engage with the original Greek text and form their own interpretations. The final three translations show noticeable improvement and precision compared to the indirect translations. Following discussions, each of the three finalized translations represents a different interpretation or approach to the text. For instance, while one of the translations is rather literal and focuses on representing the syntactic structure of the original Greek, another emphasizes lucidity and fluency. In the case of complexities that have sparked debate among translators or commentators, each Persian translation presents a unique interpretation (Shamsian and Crane 2022). The final goal is to openly publish all three translations simultaneously aligned with the original Greek to properly illustrate their variety of perspectives and approaches.

Conclusion

Based on the results of the Homeric Greek course, we conclude that cross-lingual annotations along with localized grammatical explanations provide us with an efficient and effective way of teaching ancient Greek to Persian speakers. Through mastering how to use different annotations and tools used in the Homeric course, learners were able to provide accurate translations of a Homeric Greek text, and, later on, use the same methods and techniques to understand and translate Plato's *Crito*.

Additionally, focusing on the localization of cross-lingual open resources allows us to efficiently upscale our method. We can make thousands of words of heavily annotated text available in another language simply by localizing the basic resources required for using treebanks. The grammar reference, *Didakta Grammar for Annotation*, has already proved to be easily localizable in Persian, and with minor revisions, it could be translated and localized in other languages as well.

Notes

¹ Postposition, commonly used as the marker of the object.

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About the Authors

Farnoosh Shamsian earned a BA in philosophy and an MA in Ancient Iranian Languages from the University of Tehran. She is currently a doctoral candidate at Leipzig University, focusing on teaching ancient Greek to Persian speakers using digital tools and methods of annotation. Her main research interest is the application of digital annotation such as treebanks and translation alignments in localizing pedagogical resources in low-resource languages.

Gregory Crane is a professor in the Classical Studies Department at Tufts University, an adjunct professor in the Department of Computer Science, and Editor-in-Chief of the Perseus Project. As an Alexander von Humboldt Professor, he initiated the Open Greek and Latin Project (OGLP), which aims to represent every source text produced in Classical Greek or Latin from antiquity through the present. He has long been fascinated by the interplay between rapidly advancing digital technology and the humanities.



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